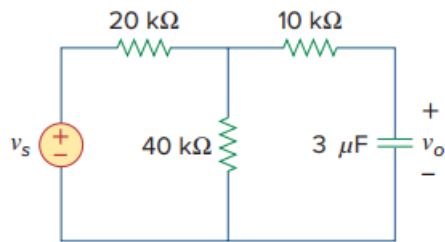


Problem

Find v_o in the circuit of Fig. 7.112 when $v_s = 30u(t)$ V. Assume that $v_o(0) = 5$ V.



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

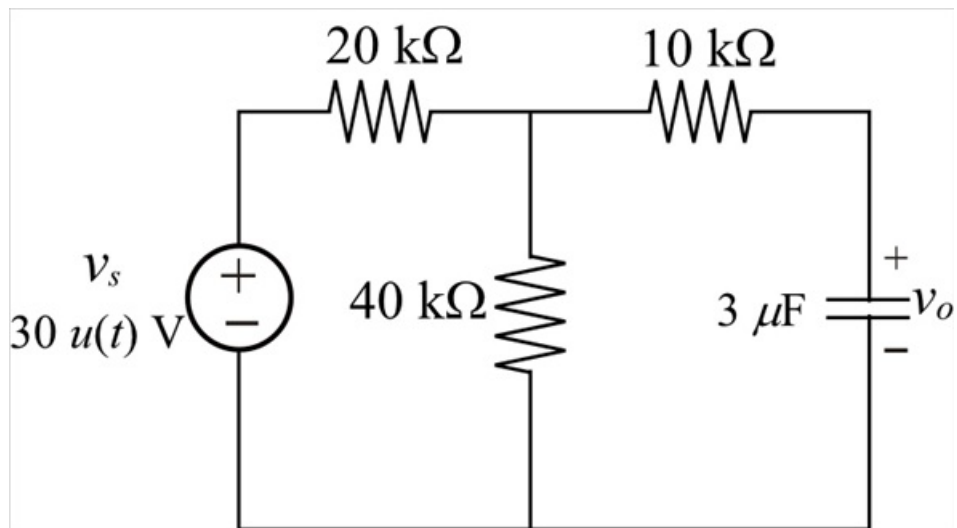


Figure 1

Step 2 of 6

Calculation of equivalent resistance:

To calculate the equivalent resistance that appears across the capacitor terminals, open the capacitor terminals and short circuit the voltage source as shown in Figure 2.

Redraw the circuit in Figure 1 to calculate the equivalent resistance:

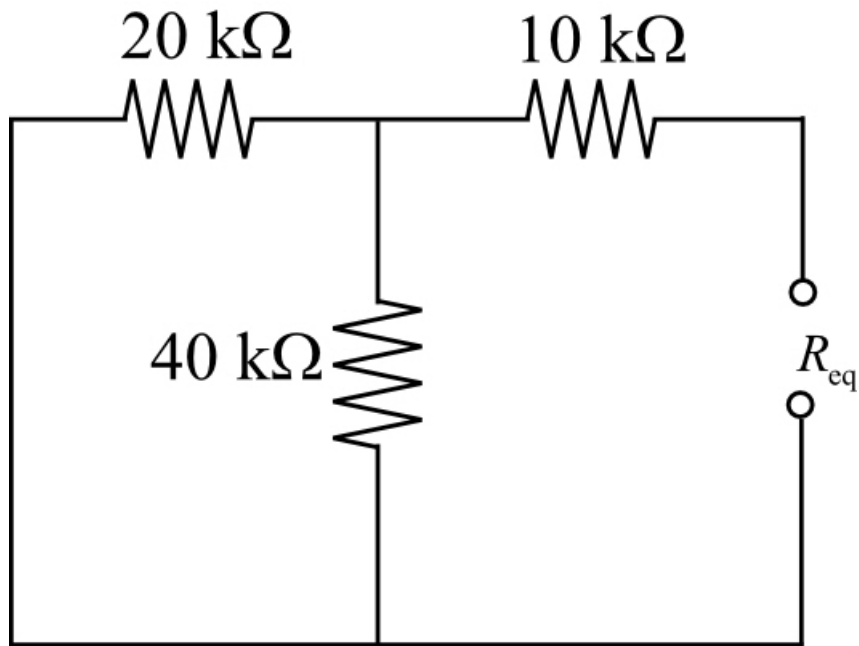


Figure 2

Step 3 of 6

From Figure 2, it is observed that both $20\text{ k}\Omega$ and $40\text{ k}\Omega$ resistors are in parallel the equivalent value and this parallel combination resistor value is in series with $10\text{ k}\Omega$.

$$\begin{aligned} R_{eq} &= (20\text{ k} \parallel 40\text{ k}) + 10\text{ k} \\ &= \left(\frac{(20\text{ k})(40\text{ k})}{20\text{ k} + 40\text{ k}} \right) + 10\text{ k} \\ &= \left(\frac{800}{60}\text{ k} \right) + 10\text{ k} \\ &= \frac{70}{3}\text{ k}\Omega \end{aligned}$$

Step 4 of 6

Redraw the circuit in Figure 1 for $t > 0$:

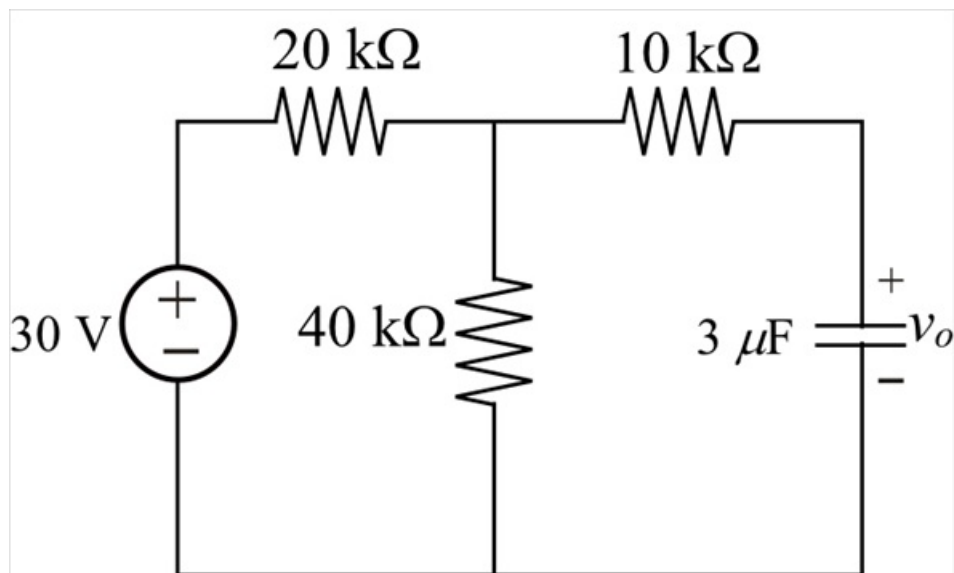


Figure 3

Step 5 of 6

After a long time, the circuit reaches steady state and the capacitor acts as an open circuit.

Redraw the circuit in Figure 3 at steady state.

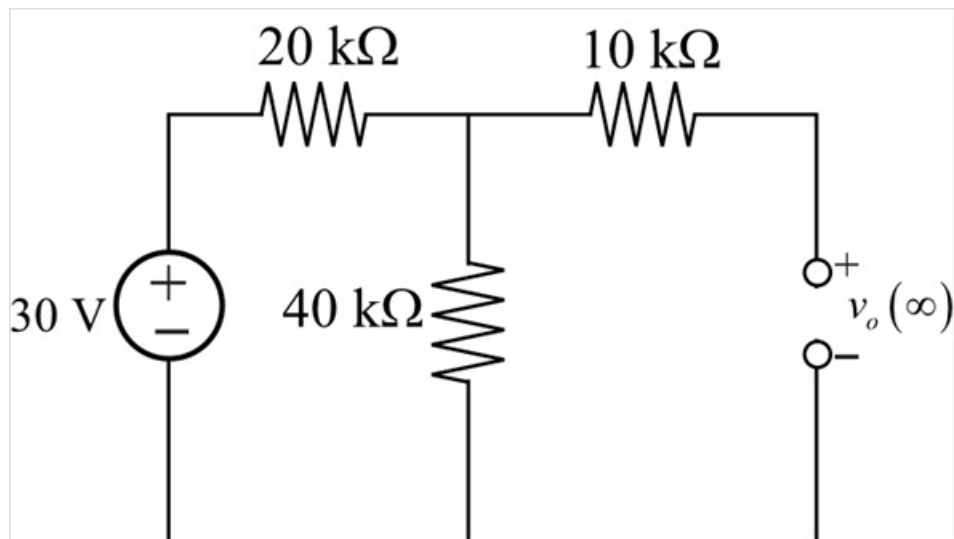


Figure 4

Step 6 of 6

Calculate $v_o(\infty)$ using voltage division formula.

$$\begin{aligned}v_o(\infty) &= \frac{40}{60}(30) \\&= \frac{2}{3}(30) \\&= 20 \text{ V}\end{aligned}$$

Calculate the time constant.

$$\begin{aligned}\tau &= R_{\text{eq}}C \\&= \left(\frac{70}{3} \times 10^3\right)(3 \times 10^{-6}) \\&= 70 \times 10^{-3} \\&= 70 \text{ msec}\end{aligned}$$

Calculate the voltage across the capacitor using the generalized expression of the step response of an RC circuit.

$$v_o(t) = v_o(\infty) + [v_o(0) - v_o(\infty)]e^{-\left(\frac{t}{\tau}\right)}$$

Substitute 5 V for $v_o(0)$, 20 for $v_o(\infty)$ and 70×10^{-3} for τ .

$$\begin{aligned}v_o(t) &= 20 + (5 - 20)e^{-\left(\frac{t}{70 \times 10^{-3}}\right)} \\&= 20 + (5 - 20)e^{-14.286t} \\&= (20 - 15e^{-14.286t}) \text{ V}\end{aligned}$$

Thus, the voltage $v_o(t)$ across the capacitor is $\boxed{(20 - 15e^{-14.286t}) \text{ V}}$.